Lab 6: Membranes — The Gatekeeper of the Cell

BIOL-1

Name: _____ **Date:** _____

Learning Objectives

By the end of this lab, you will be able to:

1. Build and label a phospholipid bilayer, identifying the hydrophilic heads and hydrophobic tails.
2. Predict which molecules cross a membrane freely, which need a channel protein, and which cannot cross.
3. Model selective permeability using a game and explain why membrane structure determines what passes.
4. Predict the direction of osmosis for cells placed in hypotonic, isotonic, and hypertonic solutions.
5. Distinguish passive transport (diffusion, osmosis, facilitated diffusion) from active transport.

Overview

Every living cell is wrapped in a plasma membrane just a few molecules thick — yet this paper-thin barrier decides everything about what enters and leaves the cell. In this lab, you will build a membrane out of paper phospholipids, run a game that tests which molecules sneak through it, and predict how cells shrink or swell in different solutions.

This lab uses the **predict → test → explain** cycle. Before each game, write a hypothesis. Then play the game, collect data, and compare the outcome to your prediction. If your prediction fails, the explanation is where the real learning happens.

No equipment is needed today — only paper and a pencil or pen. All the randomizers are made from torn paper scraps.

Materials

- 3 sheets of paper (phospholipids, randomizer slips, and molecule scraps; one doubles as the desk surface for your model)
- Pencil or pen
- Optional: a folded corner of a paper sheet, used as a draw cup/envelope to hold your randomizer slips while mixing

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Part 1: Build a Phospholipid Bilayer

1. Make your phospholipids.

On a sheet of paper, draw the template below **six times** (six phospholipids total). Draw each phospholipid inside its own box outlined with a **dashed border** — the dashed border is your tear line. Each phospholipid is drawn as a circle with two short tails hanging below it:

- The **circle** (about 3 cm across) is the hydrophilic (water-loving) **head**. Write the word **HEAD** inside it.
- The **two tails** are wavy lines, each about 4 cm long. Write **TAIL** on each one.

> ***No cutting tools needed.*** *Carefully tear along the dashed borders of each phospholipid box. Slow, steady tears work best — go corner to corner.*

2. Write your hypothesis.

Before you assemble the bilayer, answer: *If these phospholipids were dropped into water, how would they arrange themselves — and why?*

3. Assemble the bilayer.

Arrange your six torn-out phospholipids on your desk in **two rows** with the tails facing each other (heads out, tails in). **Draw the three bottom-row boxes upside-down on your template sheet (head pointing down, tails up), so when they are torn out they are already inverted** — that way the two rows meet **tails-to-tails** with heads facing the water on each side. Slide the pieces until they sit **snugly, edge to edge** — nothing to fasten, the desk holds them together. This two-row sandwich is the **phospholipid bilayer**. When you are satisfied, take the

sheet of paper beneath your model and draw a **labeled bracket or arrow** connecting the two rows, marking the membrane as one structure. Then write **"OUTSIDE the cell"** and **"INSIDE the cell"** in pencil on the desk paper beneath each side of the model — label the paper, not the phospholipid pieces.

4. Draw your model.

Draw your finished bilayer. Label the hydrophilic heads, the hydrophobic tails, the side facing outside the cell, and the side facing inside the cell.

5. Explain your assembly. Did the heads face the water or the tails? Why does this arrangement happen automatically, without any energy input? (Hint: which part of the molecule is attracted to water?)

Part 2: The Selective Permeability Game

1. Make the materials.

- **Cell boundary:** Draw a long strip on paper (about 30 cm × 5 cm) divided into three labeled sections: a wide **BILAYER** section, a narrow **CHANNEL** section, and leave one solid section labeled **BLOCKED**. Molecules will travel across this strip on your desk.
- **Molecules:** Tear out **12 small paper scraps** from a spare sheet — no precision shaping needed; different scrap sizes do the job. Label each in pencil: **3 Oxygen** (small scraps labeled **O_2**), **3 Water** (small scraps labeled **H_2O**), **3 Glucose** (large scraps, about twice the size of the oxygen scraps, labeled **glucose**), and **3 Sodium ions** (small scraps labeled **Na^+**).
- **Randomizer slips:** Fold a strip of paper into 6 squares and outline each square with a **dashed border**. Number them **1–6** in pencil, then carefully tear them apart along the dashed borders. Slow, steady tears work best — go corner to corner.

2. Write your hypothesis.

Which molecule types do you predict will cross the membrane freely through the bilayer section, which will need the channel section, and which will not cross at all? State a prediction for each of the four molecule types.

3. Game rules.

You will play **3 rounds**. In each round, draw a slip **once for each of the 12 molecules** (12 draws per round, 36 draws total). **Place all six slips face-down, mix them around, draw one, record the number, return it, and mix again.** Always return the drawn slip and mix thoroughly before the next draw — this keeps every attempt random and independent. The number on the slip tells you where that molecule attempts to cross:

- **Draw 1–2:** attempt the **BILAYER** section
- **Draw 3–4:** attempt the **CHANNEL** section
- **Draw 5:** attempt the **BLOCKED** section
- **Draw 6:** you **choose** any one section

A molecule **passes** only if it fits the section it attempts:

- **BILAYER section:** only **small nonpolar** molecules fit through the oily core. **Oxygen passes; everything else is turned away.**
- **CHANNEL section:** only **small polar or charged** molecules fit. **Water and sodium ions pass; oxygen is turned away — it does not need the channel, because it crosses the bilayer freely, and if it attempts the channel section it simply stays on the starting side this turn; glucose is too large.**
- **BLOCKED section: no molecule ever passes.**
- **Draw 6 (your choice):** apply the matching rule above for the section you choose.

Record every molecule that passes; molecules turned away are returned to the starting side. Tally the number of molecules of each type that **passed** in each round:

Molecule type	Round 1 passed	Round 2 passed	Round 3 passed
Oxygen			
Water			
Glucose			
Sodium ion (Na+)			

4. Explain your results. Which molecule(s) crossed **freely** through the bilayer, which **needed a channel**, and which **could not cross at all**? Did the outcome match your hypothesis in Step 2? If not, explain the difference.

5. Connect to the cell. Imagine the cell needs to move glucose **into** the cell against its concentration gradient. Based on your game, can it rely on the bilayer alone? What would the cell need instead — and would that process be passive or active transport?

6. Predict, then check. Which of the four molecule types would **cross easier** if the game used **6 rounds instead of 3**, and why does chance matter even when the rule is fixed?

Part 3: Osmosis and Tonicity Predictions

1. Set up the scenario.

Draw **three paper cells** — each a simple circle with a nucleus dot inside — and label them **Cell A**, **Cell B**, and **Cell C**. Each cell is placed in a different beaker of solution (imaginary — no water required!):

- **Cell A** sits in a **hypotonic** solution (fewer solutes outside than inside the cell).
- **Cell B** sits in an **isotonic** solution (solutes balanced inside and outside).
- **Cell C** sits in a **hypertonic** solution (more solutes outside than inside the cell).

2. Predict BEFORE you check.

Remember your Part 2 result: **water passed through the membrane.** Using arrows drawn on your paper cells, show which direction you predict water will move (into the cell, out of the cell, or both directions equally). Then complete each row of the table — prediction columns first, outcome columns second. **Fill in the Prediction columns before reading the outcome rules below the table.**

Cell	Solution	Prediction: water moves…	Predicted cell change	Outcome: water actually moved…	Actual cell change
A	Hypotonic				
B	Isotonic				
C	Hypertonic				

Make your two-slip randomizer: Fold a small strip of paper in half, outline it with a **dashed border** down the middle, and label one half **H** and the other **T** in pencil. Carefully tear the two slips apart along the dashed border — slow, steady tears work best, corner to corner.

Outcome rules (check your predictions): In a **hypotonic** solution, water rushes **into** the cell, which **swells and may burst (lysis)**. In an **isotonic** solution, water moves **equally both ways**, so the cell is **unchanged**. In a **hypertonic** solution, water is drawn **out** of the cell, which **shrivels (crenation)**. Draw one slip once per cell — **place both slips face-down, mix them, draw one, record it, return it, and mix again before the next cell**. Draw the **H** slip: the outcome

matches the rule exactly. Draw the **T** slip: the cell is an animal cell placed in that solution for too long, and it **ruptures in hypotonic** or **shrivels further in hypertonic**. Mark any slip-altered outcome with an asterisk in your table.

3. Explain the direction of osmosis. In your own words, why did water move the direction it did in Cells A and C? Use the words **concentration gradient** and **selectively permeable** in your answer.

4. Real-world connection. Why do hospitals dissolve IV medicines in **isotonic saline** rather than pure water? Support your answer with evidence from your Part 3 table.

Conclusion

1. One sentence synthesis. Complete this sentence: *The cell membrane is selectively permeable because…*

2. Hypothesis check. Which of your predictions from today (Part 1, 2, or 3) was furthest from the actual result, and what misconception did it reveal?

3. Looking ahead. Diffusion, osmosis, and facilitated diffusion are all **passive** transport. What single feature do they share, and how does that separate them from **active** transport like the sodium-potassium pump?